

Data Engineering Assignment

Overview

Welcome to the Basketball Data Engineering Assignment! This project is designed to evaluate your ability to model, process, and validate sports data using modern data engineering practices. You'll work with real-world basketball datasets and demonstrate your skills in data modeling, pipeline development, and quality assurance.

Provided Data Files

You are given four raw JSON files:

- `2024_NBA_Schedule.parquet` – Metadata about basketball games.
 - `2024_NBA_Players.parquet` – Player information and attributes.
 - `Boxscores/*.parquet` – Game-level player statistics.
 - `PBP/*.parquet` – Event-level data for each game.
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Part 1: Data Modeling

Objective:

Explore the raw data and design a relational data model that supports analytical queries and ensures data integrity.

Deliverables:

- Entity Relationship Diagram (ERD) or Business Requirements Document (BRD).
- Clearly define:
 - Primary Keys (PK)
 - NOT NULL constraints
 - Foreign Key relationships
 - Relationships between tables

Tools:

You may use any diagramming tool (e.g., dbdiagram.io, Lucidchart, draw.io) or markdown-based ERD formats.

⚙️ Part 2: Pipeline Implementation

Objective:

Build a batch data pipeline that ingests the raw files into a SQLite database.

Requirements:

- Use Python for orchestration and ingestion.
 - Use SQL for in-database transformations.
 - Additional thoughts:
 - Re-running the pipeline should not duplicate or corrupt data.
 - Include documentation:
 - A `README.md` or inline comments explaining how to run the pipeline.
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✅ Part 3: Data Quality & Validation

Objective:

Demonstrate your understanding of data quality principles.

Deliverables:

- A Markdown or text document describing your approach to data validation.
- Address the following:
 - Duplicate detection (e.g., repeated play-by-play events)
 - Referential integrity (e.g., player IDs in boxscores must exist in players)
 - Missing or malformed data
 - Consistency checks

🎨 Bonus (Optional): Front-End Dashboard

If time allows, build a simple dashboard using the curated SQLite data.

Ideas:

- Game Summary Viewer – Show game metadata and final scores.
- Boxscore Browser – View player stats per game.
- Player Lookup Tool – Search and display player profiles and performance.

Tools:

- Any lightweight framework (e.g., Streamlit, Flask, Dash)
- Creativity is encouraged—there are no strict requirements.

Submission Guidelines

Please submit:

- Your ERD/BRD diagram
- Python scripts and SQL files
- Data quality document
- (Optional) Dashboard code and screenshots
- A `README.md` with setup and execution instructions